Dynamic Fusion of Hyperspectral and LiDAR Data for Target Detection

Xukun Zang, Lina Xu, *Member, IEEE*, Yuxiang Zhang, *Senior Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, Qian Shi, *Senior Member, IEEE*, Lei Fang, and Bo Du, *Senior Member, IEEE*

***Abstract*—Hyperspectral target detection (HTD) has achieved significant progress by exploiting the rich spectral and spatial information. However, the spectral similarity between different types of land cover and the interference of shadows still limit the target detection performance. To address these issues, we introduce light detection and ranging (LiDAR) data to distinguish spectrally similar and shadow-affected land covers by using elevation differences. We propose a cross-modal dynamic fusion method data for target detection (DFTD) for target detection. This method constructs a cross-modal sample selection module based on the spectral information divergence (SID) and *k*-means algorithms, which applies a priori target spectra from hyperspectral image (HSI) to select target and background training samples for HSI and LiDAR data. In addition, the method develops a cross-modal feature interactive extraction (CFIE) module by a dual-branch spatial sequence mamba (DSSM) encoder and a cross-modal interactive attention module to obtain encoded features with enhanced discriminability and robustness. Furthermore, the method proposes a cross-modal dynamic feature fusion module via a designed mask deep deterministic policy gradient (MaskDDPG) algorithm, which dynamically generates modality-specific masks based on the varying discriminative capacities of HSI and LiDAR data to effectively fuse cross-modal encoded features. The fused features are fed into a detection head with fully connected layers to achieve target detection results. Extensive experiments on three publicly available HSI-LiDAR datasets demonstrate that DFTD outperforms state-of-the-art methods.**

***Index Terms*—Cross-modal fusion, deep reinforcement learning (DRL), hyperspectral image (HSI), light detection and ranging (LiDAR) data, target detection.**

I. Introduction

HYPERSPECTRAL images (HSIs) typically contain tens to hundreds of nearly continuous spectral bands, providing rich spectral and spatial information for land cover analysis [1]. With high spectral resolution, HSIs can capture subtle spectral variations to identify different types of land cover and, therefore, provide unique advantages for hyperspectral target detection (HTD) [2]. HTD has attracted increasing attention and has been widely applied in various fields such as environmental monitoring, precision agriculture, and military reconnaissance [3], [4]. The primary objective of HTD is to separate targets with known spectral signatures from various types of backgrounds, thereby achieving target enhancement and background suppression [5], [6]. Success in this task relies on effectively extracting the abundant spectral and spatial information within HSIs.

In recent years, deep learning (DL) has attracted significant attention in HTD due to its capability to learn high-level, rich features. As one of the most widely used architectures, convolutional neural networks (CNNs) have demonstrated remarkable effectiveness in extracting local spatial features with various sample expansion strategies for HTD. For example, Zhang et al. [7] proposed a deep convolutional neural network-based HTD (HTD-Net) method. This method integrates a modified autoencoder with linear prediction and pixel-pair strategies to generate training samples. Zhu et al. [8] proposed a two-stream convolutional network for target detection (TSCNTD), which combines sparse representation and linear mixing to generate abundant pseudolabeled samples. Gao et al. [9] proposed a spectral aggregation and separation network, which incorporates a target band random masking strategy to enhance discriminative features and suppress the interference caused by class imbalance in pseudolabeled data. Dong et al. [10] introduced a lightweight CNN method based on joint correlation distance constraint and density peak clustering (LCNN-CD). It obtains target samples by selecting pixels with high spectral correlation to prior targets and extracts pure background samples through density peak clustering. Building upon CNN-based approaches, recent transformer-based methods have extended feature modeling capabilities to capture long-range spectral–spatial relationships. For example, Qin et al. [11] proposed an HTD method based on a vision transformer, which constructs spectral–spatial sequences and employs a transformer encoder to jointly model each pixel and its context. Yang et al. [12] developed a compressive sensing-based triplet transformer detector (CS-TTD), which is trained with the samples generated by data augmentation using the restricted distribution property to detect targets in HSI under compressive sensing. Jiao et al. [13] proposed a triplet spectral transformer-based target detection (TSTTD) method, which uses a spectral

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Xukun Zang, Lina Xu, and Yuxiang Zhang are with the School of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China (e-mail: zangxk@cug.edu.cn; zhangyx@cug.edu.cn).

Yanni Dong is with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: dongyanni@whu.edu.cn).

Qian Shi is with Guangdong Provincial Key Laboratory for Urbanization and Geo-Simulation, School of Geography and Planning, Sun Yat-sen University, Guangzhou 510275, China (e-mail: shixi5@mail.sysu.edu.cn).

Lei Fang is with CAAZ (Zhejiang) Information Technology Company Ltd., Ningbo 315800, China (e-mail: icedark@zju.edu.cn).

Bo Du is with the School of Computer Science, Wuhan University, Wuhan 430079, China (e-mail: gunspace@163.com).

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attention mechanism combined with triplet loss and binary cross-entropy (BCE) loss to enhance target–background separability. Recently, hybrid architectures combining transformers with state space models have been explored for HTD. Zheng et al. [14] proposed a hybrid transformer-Mamba network that integrates multiscale and global–local features for background reconstruction in HTD. Furthermore, self-supervised learning-based methods exploit unlabeled data to train deep network architectures and learn general feature representations, which are useful for downstream HTD tasks. For example, Chen et al. [15] introduced a generative self-supervised target detection framework with spectral–spatial masking (GSSTD), where a lightweight vision transformer is pretrained to extract transferable features from unlabeled data. Zhang et al. [16] proposed a self-supervised target detection method based on deep clustering (SSDCTD), which conducts clustering-based pretraining on unlabeled data and guides spectral feature learning without relying on spectral variation priors.

The abovementioned DL-based HTD methods have been advanced by exploiting the rich spectral and spatial information of HSI. However, the spectral similarity between different types of land cover and the interference of shadows still limit the target detection performance. Due to factors such as land cover composition, mixed pixels, and imaging conditions, HSIs often suffer from the issue that different materials present similar or nearly identical spectral signatures. This spectral similarity usually leads to poor target enhancement and background suppression, resulting in limited target detection performance. Furthermore, as a passive sensor, HSI is susceptible to the interference of shadows, leading to degraded spectral information and weakened detection performance. To address these issues, we introduce light detection and ranging (LiDAR) data to distinguish spectrally similar and shadow-interfered land covers by using elevation differences. In scenarios where targets and backgrounds exhibit similar spectral signatures, LiDAR data can provide additional elevation information [17], [18] to assist in separating targets from backgrounds. Similarly, in shadowed areas where spectral information from HSI becomes degraded, LiDAR, as an active sensor, remains less affected by shadows and, thereby, compensates for HSI's limitations in such conditions [19], [20], [21]. However, the fusion of heterogeneous data from different sources has always been a challenge [22]. Conventional fusion methods, such as channel concatenation, fixed-weight summation, and simple attention mechanisms, typically assume uniform fusion weights of modality-specific features across different local regions [23]. Some studies have introduced interactive attention mechanisms [24], [25] to capture the relationships between modality-specific features and generate joint semantic representations. However, these fusion methods cannot dynamically adjust fusion weights based on the varying discriminative capacities of two modalities across different local regions, which results in limited exploration of cross-modal complementarity.

To address the aforementioned issue, we draw inspiration from deep reinforcement learning (DRL) and propose a cross-modal dynamic fusion method for target detection, named dynamic fusion of hyperspectral and LiDAR data for target detection (DFTD). The main idea of DFTD is to leverage feedback from DRL to dynamically allocate the fusion weights of HSI and LiDAR according to their discriminative capacities across different spatial locations. The framework consists of three parts: cross-modal sample selection, cross-modal feature interactive extraction (CFIE), and cross-modal dynamic feature fusion (CDF). First, the cross-modal sample selection module is constructed based on the spectral information divergence (SID) and k-means algorithms, which applies a priori target spectra from HSI to select target and background training samples for HSI and LiDAR data. Second, the CFIE module is designed to extract modality-specific features and cross-modal complementary information to enhance the discriminability and robustness of encoded features. The CFIE module constructs a dual-branch spatial sequence Mamba (DSSM) encoder to extract local features and capture long-range dependencies from HSI and LiDAR data, respectively. The CFIE further employs a cross-modal interactive attention module to leverage cross-modal complementary information to obtain more discriminative and robust encoded features. Then, the CDF module is proposed based on the DRL technique via the designed mask deep deterministic policy gradient (MaskDDPG) method, which fuses the encoded features of both modalities by further exploring the cross-modal complementary information from the perspective of discriminative capacities. Unlike traditional fusion methods that assume uniform fusion weights of modality-specific features across different spatial locations, the CDF module designs a MaskDDPG method via an actor–critic framework and a detection-guided reward function to dynamically generate modality-specific masks for HSI and LiDAR encoded features based on their varying discriminative capacities. Specifically, the detection-guided reward function quantifies each modality's discriminative capacity through target–background separability and detection confidence metrics, enabling the CDF module to allocate different masks for the encoded features of two modalities across different spatial locations. Finally, the fused features are processed by a detection head with fully connected layers to achieve the target detection result.

The main contributions are summarized as follows.

1) A novel DFTD target detection method is proposed based on HSI and LiDAR data, which can integrate spectral–spatial information from HSI and complementary topographic information from LiDAR data to address the spectral similarity and shadow interference issues, thereby improving the target detection performance.

2) A CFIE module is constructed by a DSSM encoder and a cross-modal interactive attention module, which can effectively obtain encoded features with enhanced discriminability and robustness.

3) A CDF module is proposed via a designed MaskDDPG method, which can dynamically generate modality-specific masks based on the varying discriminative capacities of two modalities across different spatial locations to effectively fuse cross-modal encoded features.

The rest of this article is organized as follows. Section II briefly reviews the related work. Section III describes the

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proposed DFTD method in detail. Section IV analyzes and discusses the experimental results. Finally, the conclusions are presented in Section V.

II. Related Work

A. Cross-Modal Feature Fusion

Hyperspectral and LiDAR data capture details of the same land cover scene from different perspectives for image interpretation tasks [26], [27]. In the field of hyperspectral and LiDAR joint classification, many fusion methods have been proposed to leverage the cross-modal complementarity to improve classification performance [29], [30], [31]. These methods can be classified into three types: pixel-level fusion, feature-level fusion, and decision-level fusion [28], [32].

The pixel-level fusion approach directly utilizes raw data by integrating cross-modal images at the pixel scale, typically employing conventional techniques such as channel concatenation or weighted averaging [33]. However, simple image merging approaches may fail to fully exploit the distinctive features of each modality while incurring high computational complexity. The decision-level fusion approach integrates the classification results of each modality, commonly employing statistical rules or weighted averaging to formulate the final classification decision [34]. Despite its operational simplicity and computational efficiency, this approach may inadequately capture intermodal synergistic information. In contrast, the feature-level fusion approach extracts features from each modality and merges them into a unified representation [35], [39]. Due to its ability to effectively integrate cross-modal information [36], [37], this approach has been widely used in cross-modal data fusion [38] and has become a key research direction [40].

The key challenge of feature-level fusion lies in effectively capturing complex cross-modal interactions [41]. Especially when handling heterogeneous data, cross-modal correlations and complementarities are often not fully explored. This limits the model's complete understanding of cross-modal data. To overcome this limitation, cross-modal feature interaction has emerged as a research hotspot. This paradigm projects features from different modalities into a unified space and models their complementary relationships, thereby unlocking the latent potential of cross-modal data [42]. For example, Lin et al. [43] proposed a dynamic cross-modal feature interaction network, which enhances the interaction between HSI and LiDAR features through bilinear spatial and channel attention blocks, combined with a dynamic routing mechanism to optimize the feature fusion path. Zhang et al. [44] proposed a local-to-global cross-modal attention fusion framework, which captures local and global features from HSI and LiDAR data using a cross-attention mechanism. Cai et al. [45] proposed a graph-attention-based multimodal fusion network. This network represents HSI and LiDAR features using an undirected weighted graph and applies a graph attention strategy to fuse node information and explore relationships between single-source and multisource data.

Although these methods perform well in the field of hyperspectral and LiDAR joint classification, the integration of hyperspectral and LiDAR data fusion for target detection remains underexplored to the best of our knowledge. Inspired by the above feature fusion methods, we design a CFIE module for target detection.

B. Deep RL

Reinforcement learning (RL) is a machine learning paradigm in which an agent learns the optimal action through repeated interaction with the environment, aiming to maximize cumulative reward [46]. The agent is guided by a reward function to act toward goal-oriented strategies [47]. DRL extends RL by combining deep neural networks, enhancing its ability to handle high-dimensional state spaces [48]. This combination allows DRL algorithms to solve complex decision-making tasks without explicit models. DRL has been applied in various fields such as robotics, autonomous driving, and remote sensing [49]. For example, Guo et al. [50] proposed a two-stage unsupervised hyperspectral band selection method based on DRL. The method models the band selection problem as a Markov decision process, where the agent learns an optimal band selection strategy by interacting with an environment. Cai et al. [51] proposed a method called RL from AI feedback to enhance building information extraction in remote sensing images. By integrating AI feedback into the RL framework, they employed a custom reward network and a deep multiscale attention mechanism. This significantly reduced pixelwise cross-entropy loss and improved recall and accuracy. Lyu et al. [52] proposed a composite control strategy for quadruped robots by combining DRL with a model-based controller. By generating gait commands using DRL and correcting them with a model-based controller based on the system's dynamic model, they enhanced the robot's adaptability and robustness in complex terrains. Wu et al. [53] introduced an end-to-end policy learning method based on DRL to address challenges of sparse rewards and exploration in autonomous driving scenarios. They reduced overestimation bias in the value function by introducing intrinsic reward signals based on recurrent neural networks and an auxiliary critic network.

Deep deterministic policy gradient (DDPG) is a typical DRL algorithm that combines policy gradient and Q-learning [54]. It is designed for continuous action spaces and follows the actor–critic framework. The actor network generates deterministic action for a given state, while the critic network evaluates the Q value of state–action pair. The algorithm trains two networks to optimize the deterministic policy and value function. The critic network updates its parameters by minimizing the difference between predicted Q-values and target Q-values, and the actor network optimizes the policy by maximizing the Q value output from the critic network. To stabilize training, DDPG employs experience replay and target network mechanisms. To guide the training direction, DDPG employs a reward function. The deterministic policy of DDPG is highly compatible with the HSI-LiDAR joint target detection task. For a given cross-modal patch pair, the actor network of DDPG can generate a deterministic mask pair to modulate the fusion weights of the two modalities. This deterministic property ensures the stability and reproducibility of target detection results.

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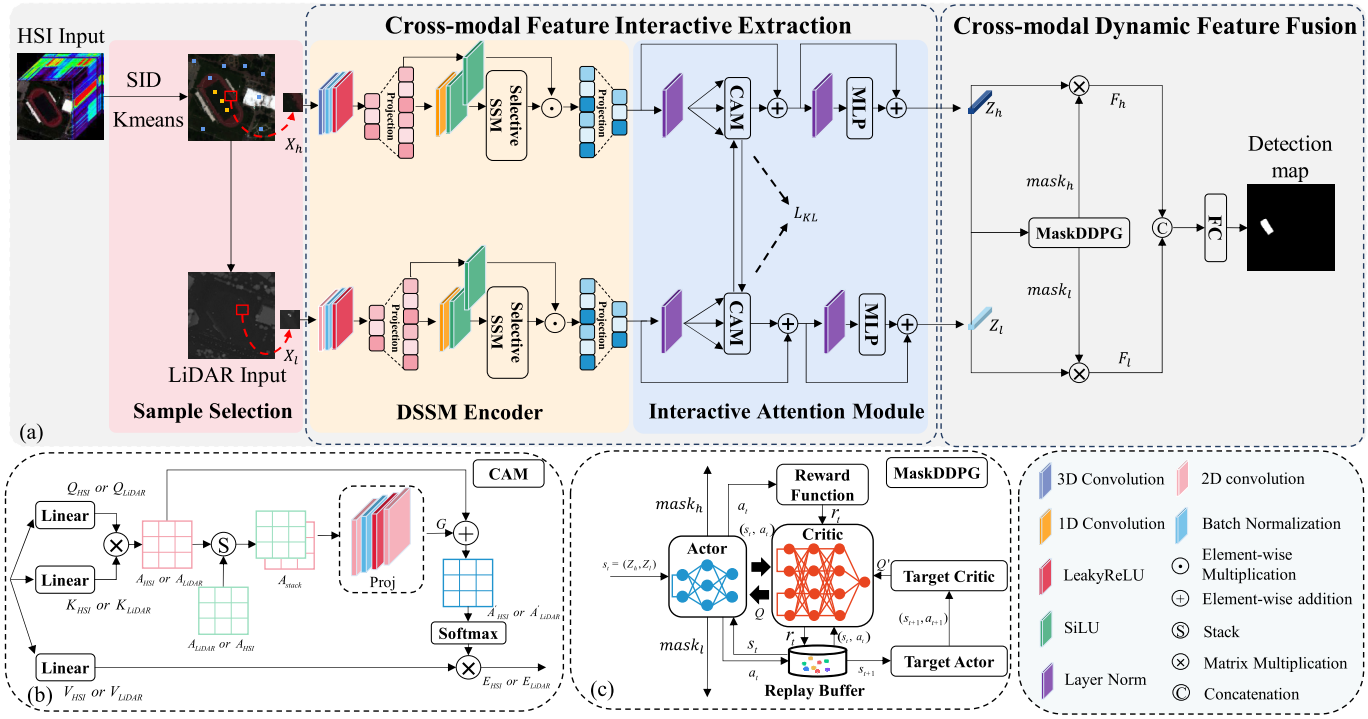


Fig. 1. Architecture of the proposed DFTD method. (a) Overall framework. (b) CAM. (c) MaskDDPG.

III. Method

The proposed method can be divided into three parts: cross-modal sample selection, CFIE, and cross-modal dynamic feature fusion. The overall architecture of DFTD is illustrated in Fig. 1.

A. Cross-Modal Sample Selection

The first key step in hyperspectral and LiDAR joint target detection is to select target and background training samples from cross-modal data. How to conduct the cross-modal sample selection deserves further consideration since there are only a few priori target spectra for HSI, and there is no a priori information in LiDAR data. Based on the characteristic that cross-modal data possess identical spatial resolution after undergoing necessary preprocessing prior to the target detection task, we design a straightforward yet effective cross-modal sample selection method. Specifically, this method first employs a priori target spectra to select training samples from HSI and then leverages pixel-to-pixel correspondence to extract corresponding training samples from the LiDAR data.

First, the SID algorithm is employed to select the pixels similar to a priori target spectra as target samples for HSI. For each pixel, the SID values with all priori target spectra are computed, and the minimum SID value is selected as the final SID value for this pixel. Then, pixels with the M smallest SID values are selected as target samples, where M equals the number of a priori target spectra. The output of the SID algorithm can be represented as follows:

$$\mathrm{SID}(i) = \min_{k=1,\ldots,M} \frac{1}{2}\left(D_{\mathrm{KL}}(S_i \| T_k) + D_{\mathrm{KL}}(T_k \| S_i)\right) \tag{1}$$

where S_i and T_k denote the normalized spectral probability distributions of the ith pixel in the HSI and the kth priori target spectrum, respectively. $D_{\mathrm{KL}}(S_i \| T_k)$ is the Kullback–Leibler (KL) divergence, which measures the difference between the probability distributions S_i and T_k. $D_{\mathrm{KL}}(S_i \| T_k)$ is computed as follows:

$$D_{\mathrm{KL}}(S_i \| T_k) = \sum_j S_i(j) \log \frac{S_i(j)}{T_k(j)}. \tag{2}$$

Then, the k-means algorithm is employed to generate cluster centers. The pixel nearest each cluster center is selected as a representative background sample for HSI. The number of selected background samples is analyzed in Section IV.

Finally, the target and background samples for LiDAR data are obtained based on the pixel-to-pixel correspondence of HSI and LiDAR data. The obtained target and background training samples are assigned labels of 1 and 0, respectively.

B. Cross-Modal Feature Interactive Extraction

The CFIE module is designed to extract features from HSI and LiDAR data. The module consists of a DSSM encoder and an interactive attention module. The DSSM encoder aims to extract local features and long-range dependencies from HSI and LiDAR data using CNN and Mamba. The interactive attention mechanism aims to capture the complementary information between modality-specific features, enhancing the discriminability and the robustness of encoded features.

In the CFIE input, we first divide HSI and LiDAR data into regular nonoverlapping patches, denoted as $X_h \in \mathbb{R}^{s\times s\times c_1}$ and $X_l \in \mathbb{R}^{s\times s\times c_2}$, where $s \times s$ represents the spatial size of each patch ($s = 3$), and c_1 and c_2 are the original channel dimensions of HSI and LiDAR, respectively.

The DSSM encoder includes an HSI branch and a LiDAR branch. Two branches consist of two different convolutional blocks and two same Mamba blocks. For the HSI branch,

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the convolution block consists of a 3-D convolution layer, a batch normalization layer, and a leaky rectified linear unit (LeakyReLU) function layer. The 3-D convolution kernel size is set as $1 \times 1 \times 7$, and the stride along the spectral dimension is set as 2. This convolution block halves the spectral dimension to remove redundant spectral information and extracts the local spectral features of HSI data. For the LiDAR branch, the convolution block consists of a 2-D convolution layer with a kernel size of 3×3, a batch normalization layer, and a LeakyReLU layer. This convolution block explores the local spatial features of the LiDAR data. For both the HSI and LiDAR branches, the Mamba block leverages a selective state space model with input-dependent parameters to effectively extract long-range dependencies while maintaining linear computational complexity.

The interactive attention module aims to capture complementary information between HSI and LiDAR features through a dual-residual architecture. The module employs a cross-modal attention mechanism (CAM) and a multilayer perceptron (MLP) with residual connections to enhance the discriminability and robustness of encoded features. The CAM is illustrated in Fig. 1(b).

Specifically, CAM first independently calculates the attention logits for HSI and LiDAR modalities

$$A_{HSI} = \frac{Q_{HSI} K_{HSI}^{\mathrm{T}}}{\sqrt{d_k}} \tag{3}$$

$$A_{LiDAR} = \frac{Q_{LiDAR} K_{LiDAR}^{\mathrm{T}}}{\sqrt{d_k}} \tag{4}$$

where Q_{HSI} and Q_{LiDAR} are the query matrices. K_{HSI} and K_{LiDAR} are the key matrices. d_k is the dimension of the key, which is set to be identical for both HSI and LiDAR modalities.

Subsequently, the two attention logits are stacked along a new dimension to form a tensor $A_{stack} \in \mathbb{R}^{b \times h \times 2 \times n \times n}$, where b is the batch size, h is the number of attention heads, and n is the sequence length. A projection module Proj is then applied to learn the interaction between the attention maps of different modalitieswhere Proj is implemented as a cascade of 1×1 convolutional layers. The output $G \in \mathbb{R}^{b \times h \times n \times n}$ represents the learned cross-modal attention supplement that captures the complementary information between HSI and LiDAR modalities. The attention logits of both modalities are then enhanced by adding the learned supplement

$$A'_{HSI} = A_{HSI} + G \tag{6}$$

$$A'_{LiDAR} = A_{LiDAR} + G \tag{7}$$

where A'_{HSI} and A'_{LiDAR} denote the enhanced attention logits for HSI and LiDAR modalities, respectively.

To encourage collaborative learning between the attention distributions of the HSI and LiDAR modalities, we introduce the KL divergence as a constraint term to compute the difference between the enhanced attention logits of the two modalities. The computation of the KL divergence loss can be expressed as follows:

$$L_{\mathrm{KL}} = D_{\mathrm{KL}}\left(p_{HSI} \parallel p_{LiDAR}\right) \tag{8}$$

where p_{HSI} and p_{LiDAR} represent the attention distributions of the HSI and LiDAR modalities after enhancement, respectively. By minimizing the KL divergence, we regularize the attention distributions of the two modalities during training, thereby enhancing the fusion of cross-modal information and promoting collaborative learning between the modalities.

Finally, the enhanced attention logits are normalized through softmax and applied to the value matrix V to obtain the cross-modal enhanced feature representations. This process allows the model to capture the global dependencies within each modality and understand the significance of each feature in its context. The cross-modal enhanced feature representations can be obtained as follows:

$$E_{HSI} = \mathrm{softmax}\left(A'_{HSI}\right) \cdot V_{HSI} \tag{9}$$

$$E_{LiDAR} = \mathrm{softmax}\left(A'_{LiDAR}\right) \cdot V_{LiDAR}. \tag{10}$$

The output features are further processed through a linear layer and an MLP to generate high-dimensional semantic feature representations. These encoded feature vectors, denoted as Z_h and Z_l for HSI and LiDAR, respectively, serve as the state input for the subsequent CDF module.

C. Cross-Modal Dynamic Feature Fusion via MaskDDPG

The CDF module via MaskDDPG is designed to effectively fuse cross-modal encoded features. The module via MaskDDPG consists of actor and critic networks, along with their target counterparts, a replay buffer, and a reward function. Its overall structure is illustrated in Fig. 1(c).

Actor Network: The actor network is designed to learn a mask generation strategy that dynamically allocates fusion weights for HSI and LiDAR encoded features based on their discriminative capacities. The actor network processes the encoded features Z_h and Z_l from HSI and LiDAR through fully connected layers with ReLU activation to generate a 2-D action vector $b = [b_1, b_2]^{\mathrm{T}}$, where $b_1, b_2 \in [-1, 1]$.

To prevent the model from converging to local optima, the action vector is added with Gaussian noise with a mean of 0 and a standard deviation of $\sigma = 0.1$, which can be expressed as follows:

$$a = b + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim N\left(0, \sigma^2 \mathbf{I}\right) \tag{11}$$

where $a = [a_1, a_2]^{\mathrm{T}}$ is the noise-perturbed action vector with a_1 and a_2 being its two components. These components are then transformed into mask values constrained to the range [0,1] using the following formula:

$$mask_h = 0.5 \times (1 + a_1) \tag{12}$$

$$\begin{aligned} G &= \mathrm{Proj}\left(\mathrm{Stack}\left[A_{HSI}, A_{LiDAR}\right]\right) \\ &= \mathrm{Conv}_{1\times1}\left(\mathrm{LeakyReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Stack}\left[A_{HSI}, A_{LiDAR}\right]\right)\right)\right)\right) \end{aligned} \tag{5}$$

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$$mask_l = 0.5 \times (1 + a_2) \tag{13}$$

where $mask_h$ and $mask_l$ represent the mask values for HSI and LiDAR encoded features, respectively. The mask values are expected to quantify the discriminative capacity of each modality, which is implemented by the following designed detection-guided reward function. For the patches across different spatial locations, the actor network generates different mask values for the encoded features from HSI and LiDAR data. In this way, the CDF module obtains dynamic weights to modulate its contributions in the feature fusion process.

Critic Network: The role of the critic network is to evaluate the effectiveness of the mask strategy generated by the actor network. The critic network calculates the Q value of the current mask and provides feedback to the actor network. The goal of the critic network is to minimize the error in the Q value prediction, thereby improving the mask generation strategy evaluation and guiding the actor's mask generation strategy enhancement. The Q value calculation follows the Bellman equation and can be represented as follows:

$$Q_{\text{target}} = r_t + \gamma \cdot Q'\left(s_{t+1}, a_{t+1}\right) \tag{14}$$

where Q_{target} is the target Q value, representing the expected return for the current mask. r_t is the immediate reward calculated through the reward function. γ is the discount factor for controlling the impact of future rewards. $Q'(s_{t+1}, a_{t+1})$ is the expected return computed by the target critic network, where the next action a_{t+1} is produced by the target actor network based on the next state s_{t+1}.

The error L_{critic} between the current predicted Q value and the target Q value is calculated to optimize the network

$$L_{\text{critic}} = \mathbb{E}\left[\left(Q_{\text{target}} - Q\left(s_t, a_t\right)\right)^2\right] \tag{15}$$

where $Q(s_t, a_t)$ is the predicted Q value computed using the critic network.

The actor network updates its parameters by maximizing the Q value computed by the critic network to improve the mask generation strategy

$$L_{\text{actor}} = -\mathbb{E}\left[Q\left(s_t, a_t\right)\right]. \tag{16}$$

It should be noted that L_{actor} and L_{critic} are used to update the actor and critic networks within the MaskDDPG method. During the training of the main DFTD detection network, the actor–critic framework performs multiple internal updates in each epoch to optimize the mask generation strategy.

Target Networks: To stabilize the DDPG training process, we introduce a target network mechanism. This mechanism includes a target actor network and a target critic network. The target networks share the same architecture as the actor–critic networks but maintain independent parameters. During training, the target networks are used to compute the target Q value, and this separated network design prevents instability in Q value estimation. The parameters of target networks are updated through a soft update mechanism rather than direct copying from the actor–critic networks. The parameters are updated as follows:

$$\theta' \leftarrow \rho\theta + (1 - \rho)\,\theta' \tag{17}$$

$$\phi' \leftarrow \rho\phi + (1 - \rho)\,\phi' \tag{18}$$

where θ and ϕ denote the parameters of the actor and critic networks, respectively; θ' and ϕ' denote the parameters of the target actor and critic networks, respectively; and ρ is the soft update coefficient, controlling the rate at which target network parameters approach the actor–critic network parameters.

Replay Buffer: The replay buffer stores experience data generated during the training process, including states, actions, rewards, and next states. Furthermore, it breaks the correlation in the data by sampling them randomly for experience replay. The random sampling approach helps improve the stability of training and accelerates the convergence of the mask generation policy. The replay buffer stores each state $s_t = (Z_h, Z_l)$, action a_t, reward r_t, and next state s_{t+1}. In each training batch, a minibatch is randomly sampled from the replay buffer to perform 15 iterative updates of the actor and critic networks before the backpropagation of the main network.

During training, the actor network generates action a_t based on the current state $s_t = (Z_h, Z_l)$ and converts it to masking probabilities, which are applied to HSI and LiDAR feature modulation. The masked features are fed forward through the network to obtain the next state s_{t+1}, while the immediate reward r_t is calculated using the reward function. The experience tuple (s_t, a_t, r_t, s_{t+1}) is stored in the replay buffer. During network updates, minibatches $\{(s_t, a_t, r_t, s_{t+1})\}$ are sampled from the buffer for optimization. The state s_t and the action a_t are fed into the critic network to compute $Q(s_t, a_t)$; simultaneously, s_t is also fed into the actor network to generate new actions for strategy optimization. The next state s_{t+1} is fed into the target actor network to generate a_{t+1}, and then, (s_{t+1}, a_{t+1}) is fed into the target critic network to compute $Q'(s_{t+1}, a_{t+1})$ for computing the target Q value.

Reward Function: The reward function of MaskDDPG is designed as a composite function, consisting of two components: separation reward and confidence reward.

The separation reward is designed to evaluate the discriminative ability of the encoded feature representations in distinguishing the target from the background. It quantifies the degree of feature separation by computing the ratio of interclass distance to intraclass distance between target and background feature vectors. Specifically, it encourages the model to produce features with large interclass distances and penalizes dispersed intraclass features, which is expected to improve the performance of target–background separation and background suppression. The separation reward is calculated as follows:

$$R_{sep} = \frac{D_{HSI} + D_{LiDAR}}{D_{HSI}^{\text{intra}} + D_{LiDAR}^{\text{intra}}} \tag{19}$$

where D_{HSI} and D_{LiDAR} denote the average Euclidean distances between the target and background classes in the HSI and LiDAR feature spaces, respectively. D_{HSI}^{intra} and D_{LiDAR}^{intra} represent the sum of average intraclass distances for the target and background classes in the HSI and LiDAR feature spaces, respectively.

The confidence reward is designed to encourage the model to improve the detection ability of the target class based on the masked encoded features. This reward is implemented

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by calculating the probability of the target. The reward is computed by applying the sigmoid activation function to convert the predicted value of the target class into a probability, as shown in the following formula:

$$R_{conf} = \frac{1}{N_{\text{target}}} \sum_{i=1}^{N_{\text{target}}} p_{\text{target}}^{(i)} \tag{20}$$

where $p_{\text{target}}^{(i)}$ is the predicted probability of the ith target class sample by the model and N_{target} represents the total number of target class samples in the current batch.

The total reward is the weighted sum of the separation reward and confidence reward, which can be represented as follows:

$$R_{total} = R_{sep} + \lambda_1 R_{conf} \tag{21}$$

where λ_1 is the weight of confidence reward, which is used to manage the tradeoff between two reward components.

Guided by the detection-oriented reward function, the CDF module dynamically adjusts the mask values to weight the contributions of different modalities based on their discriminative capacities. The mask values are applied elementwise to the feature vectors through scalar multiplication

$$F_h = Z_h \times mask_h \tag{22}$$

$$F_l = Z_l \times mask_l \tag{23}$$

where F_h represents the masked HSI features and F_l represents the masked LiDAR features. They are fused for final detection via the following formula:

$$F_{fuse} = \text{Concatenate}\left(F_h, F_l\right). \tag{24}$$

The fused feature vector is then passed through a fully connected layer followed by sigmoid activation to produce the detection probability

$$y = \text{sigmoid}\left(\text{FC}\left(F_{fuse}\right)\right) \tag{25}$$

where FC$(\cdot)$ denotes the fully connected layer and $y \in [0, 1]$ represents the probability that a pixel belongs to the target.

Since the HTD task is essentially a binary classification problem, we use the BCE loss function to train the DFTD model. This loss function is used to measure the difference between the predicted probability of the target class and the actual label, which can be calculated as follows:

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log\left(p_i\right) + \left(1 - y_i\right) \log\left(1 - p_i\right)\right] \tag{26}$$

where N is the number of samples, y_i is the actual label of the ith sample, and p_i is the predicted probability of the ith sample being the target. The overall loss of the DFTD method is the sum of BCE loss and the KL divergence loss, which can be represented as follows:

$$L_{\text{total}} = L_{\text{BCE}} + \lambda_2 \cdot L_{\text{KL}} \tag{27}$$

where λ_2 is the weight of the KL divergence loss, which is used to manage the tradeoff between two loss terms.

IV. Experiment

A. Experiment Details

1) Dataset Description: This study uses three publicly available HSI and LiDAR datasets to evaluate the performance of DFTD. The priori target spectra are obtained from three pixels near the geometric center of the target objects.

Houston2013 Dataset [55]: The Houston2013 dataset was collected on the University of Houston campus and the surrounding urban area. Both the HSI and LiDAR data consist of 349×1905 pixels. The HSI data contain 144 bands, spanning the wavelength range of 380–1050 nm. In this study, synthetic grass is regarded as the target of interest, covering 0.10% of the entire image. This dataset contains some shadowed areas.

Muufl Dataset [56]: The Muufl dataset was collected on the campus of the University of Southern Mississippi, consisting of both HSI and LiDAR data. The HSI data include 64 usable bands, with a wavelength range from 375 to 1050 nm. Both the HSI and LiDAR data have a spatial size of 325×220 pixels. In this study, water is regarded as the target of interest, covering 0.95% of the entire image.

SZUTreeData-R2 [57], [58], [59], [60]: The SZUTreeData-R2 dataset was collected on the Yuehai campus of Shenzhen University. After removing noisy bands, the HSI data consist of 98 bands, with a wavelength range from 399 to 924 nm. Both the HSI and LiDAR data have a size of 4040×2444 pixels. In this study, a region of 1000×1000 pixels is selected as the area of interest, with *Plumeria rubra* as the target of interest, covering 0.67% of the entire image.

2) Experimental Settings: To evaluate the performance of the proposed DFTD method, we select six DL-based detection methods for comparison experiments, including LCNN-CD [10], SSDCTD [16], TSTTD [13], OS-VAE [61], GSSTD [15], and CS-TTD [12]. All comparison methods are implemented using the best parameter settings recommended in the original papers. The comparison methods use HSI as input, while the proposed method uses both HSI and LiDAR as input.

The hardware configuration used in the experiments is an NVIDIA GeForce RTX 3090 graphics processing unit. We implement the proposed method using the PyTorch [62] DL framework. All trainable parameters of the neural networks in the DFTD method are randomly initialized, and no pretrained weights are used. The learning rate, the number of epochs, and the batch size are set to $2e^{-4}$, 300, and 256, respectively. The AdamW optimizer is used.

3) Evaluation Criterion: To comprehensively evaluate detection performance, multiple evaluation metrics are employed in this study, including detection maps, the receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC), separability maps, and runtime. By segmenting the detection maps through the adjustment of various thresholds (τ), the corresponding detection rate (D) and the false alarm rate (F) [63] are obtained, enabling the plotting of the ROC curve. The horizontal axis of the ROC curve represents the false alarm rate (F), and the vertical axis represents the detection rate (D). The closer the curve is to the upper left corner, the better the detection performance.

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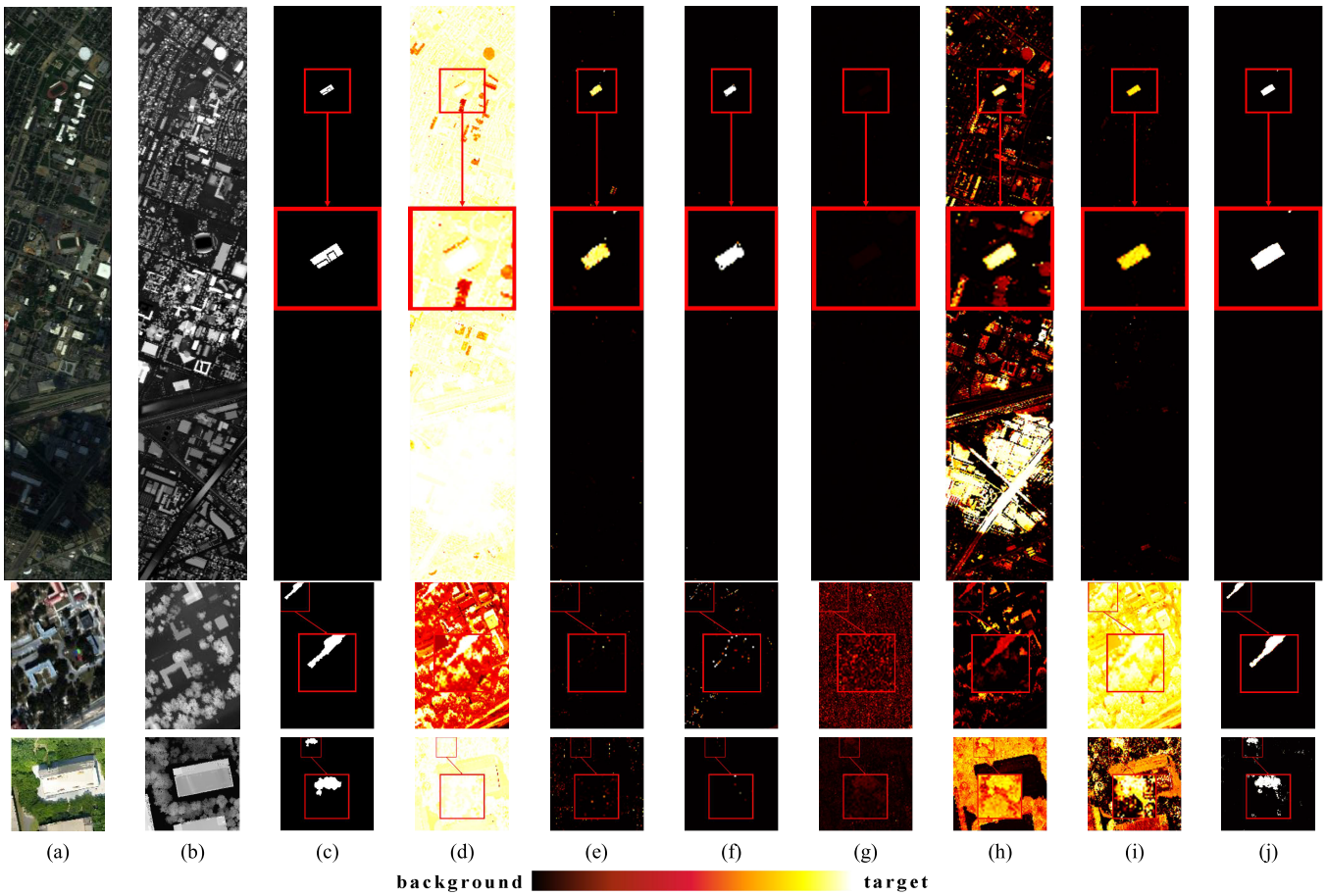


Fig. 2. Detection maps of different methods. (a) Pseudocolor images of Houston2013, Muufl, and SZUTreeData-R2 datasets. (b) Gray-scale image of LiDAR. (c) Ground truth. (d) LCNN-CD. (e) SSDCTD. (f) TSTTD. (g) OS-VAE. (h) GSSTD. (i) CS-TTD. (j) DFTD.

A series of AUC variants is constructed based on the variables D, F, and τ [64], [65]. The classical $\mathrm{AUC}_{(F,D)}$ represents the area under the ROC curve of F and D, with a larger area indicating better performance of the detector. $\mathrm{AUC}_{(\tau,F)}$ represents the area under the ROC curve of τ and F, where a smaller area indicates better background suppression performance. $\mathrm{AUC}_{(\tau,D)}$ represents the area under the ROC curve of τ and D, with a larger area indicating better target detection performance. To comprehensively evaluate the detector's performance, we adopt three additional AUC variants proposed in prior work. Specifically, $\mathrm{AUC}_{\mathrm{TD}}$ is the sum of $\mathrm{AUC}_{(F,D)}$ and $\mathrm{AUC}_{(\tau,D)}$, mainly reflecting the target detection performance; $\mathrm{AUC}_{\mathrm{BS}}$ is the difference between $\mathrm{AUC}_{(F,D)}$ and $\mathrm{AUC}_{(\tau,F)}$, further measuring the background suppression performance; and $\mathrm{AUC}_{\mathrm{ODP}}$ is the sum of $\mathrm{AUC}_{(\tau,D)}$ and $(1 - \mathrm{AUC}_{(\tau,F)})$, providing a comprehensive evaluation of both target detection and background suppression performance.

The separability map is an effective visual tool that assesses the separability of targets and backgrounds [66]. By examining the degree of separation in the map, it provides an intuitive evaluation of the method's performance. Furthermore, the runtime of the model serves as a key evaluation metric, measuring its computational efficiency.

B. Detection Performance

This section provides a detailed analysis of the detection performance of the proposed DFTD method by comparing it with several detection methods.

As illustrated in Fig. 2, the detection maps demonstrate that the DRLDF method achieves superior target detection performance across all three datasets, which depicts target contours clearly and exhibits strong background suppression capabilities. In contrast, the comparison methods such as LCNN-CD, GSSTD, and CS-TTD exhibit limited detection performance, characterized by blurred target edges and substantial background interference. LCNN-CD and GSSTD exhibit obvious false alarms in shadow areas. The comparison methods, such as SSDCTD, TSTTD, and OS-VAE, show some degree of background suppression in three datasets but fail to fully detect the targets in Muufl and SZUTreeData-R2 datasets.

The ROC curves of different methods on the three datasets are shown in Fig. 3. The ROC curves of DFTD are consistently the closest to the top-left corner across all three datasets and significantly far away from those of all comparison methods on Muufl and SZUTreeData-R2 datasets. On the Houston2013 dataset, the ROC curves of CS-TTD and TSTTD are close to DFTD when the false alarm rate is larger than 10^{-4}, showing competitive performance.

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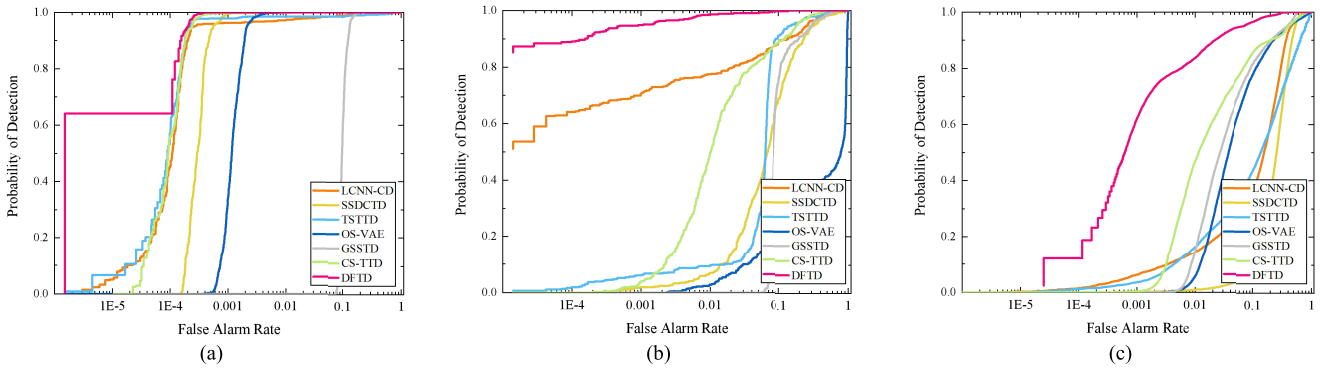


Fig. 3. ROC curves of different methods on three datasets. (a) Houston2013, (b) Muufl, and (c) SZUTreeData-R2.

TABLE I
AUC VALUES OF DIFFERENT METHODS FOR THREE DATASETS

		LCNN-CD	SSDCTD	TSTTD	OS-VAE	GSSTD	CS-TTD	DFTD
Houston2013	$AUC_{(F,D)}$	0.9974	0.9997	0.9937	0.9987	0.8981	0.9999	**0.9999**
	$AUC_{(\tau,F)}$	0.1814	0.0013	**0.0003**	0.0006	0.1831	0.0012	0.0004
	$AUC_{(\tau,D)}$	0.1896	0.8362	0.9749	0.0068	0.8923	0.7115	**0.9976**
	AUC_{TD}	1.1870	1.8358	1.9686	1.0056	1.7905	1.7114	**1.9975**
	AUC_{BS}	0.8161	0.9983	0.9934	0.9981	0.7150	0.9987	**0.9995**
	AUC_{ODP}	1.0082	1.8348	1.9746	1.0062	1.7092	1.7103	**1.9973**
Muufl	$AUC_{(F,D)}$	0.9620	0.8975	0.9295	0.4231	0.8854	0.9629	**0.9990**
	$AUC_{(\tau,F)}$	0.1866	0.0019	0.0053	0.0108	0.0635	0.8030	**0.0001**
	$AUC_{(\tau,D)}$	0.3564	0.0143	0.0803	0.0124	0.2608	**0.9780**	0.8924
	AUC_{TD}	1.3184	0.9118	1.0098	0.4355	1.1462	**1.9409**	1.8914
	AUC_{BS}	0.7754	0.8957	0.9242	0.4122	0.8220	0.1598	**0.9989**
	AUC_{ODP}	1.1698	1.0124	1.0750	1.0015	1.1973	1.1749	**1.8923**
SZUTreeData-R2	$AUC_{(F,D)}$	0.8201	0.7286	0.7637	0.9068	0.9260	0.9325	**0.9886**
	$AUC_{(\tau,F)}$	0.0440	0.0083	0.0002	**0.0000**	0.3409	0.3091	0.0032
	$AUC_{(\tau,D)}$	0.0457	0.0088	0.0154	0.0000	0.6982	**0.8605**	0.7702
	AUC_{TD}	0.8658	0.7374	0.7792	0.9068	1.6242	**1.7930**	1.7588
	AUC_{BS}	0.7761	0.7203	0.7635	0.9068	0.5851	0.6235	**0.9854**
	AUC_{ODP}	1.0017	1.0005	1.0152	1.0000	1.3573	1.5514	**1.7670**

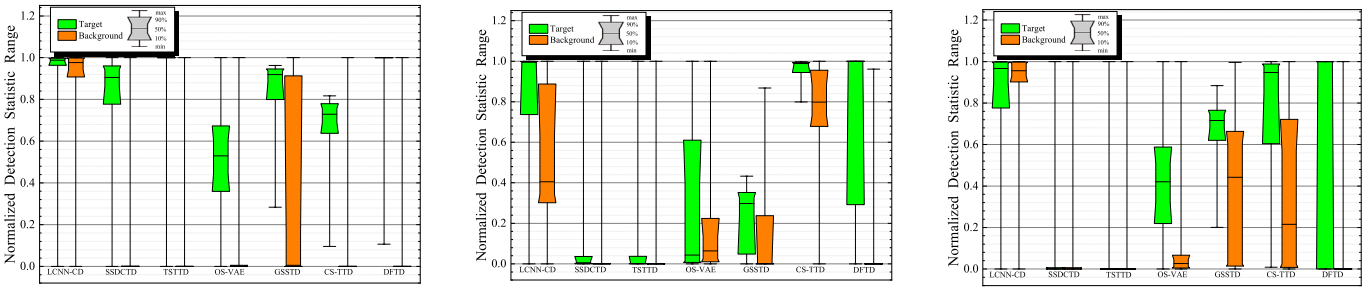


Fig. 4. Separability maps of different methods on three datasets. (a) Houston2013, (b) Muufl, and (c) SZUTreeData-R2.

Table I presents the AUC values obtained by different methods over the three datasets, with the highest value highlighted in bold. Overall, DFTD performs the best on over half of the evaluation criteria. In detail, DFTD obtains the highest $AUC_{(F,D)}$ values on all three datasets, validating its superior performance in target detection tasks. In addition, DFTD achieves the highest AUC_{BS} scores on all datasets, further demonstrating its robust background suppression performance. Notably, DFTD ranks second on $AUC_{(\tau,D)}$ and AUC_{TD} metrics on Muufl and SZUTreeData-R2 datasets, while DFTD obtains the highest AUC_{ODP} values across all datasets, indicating that the method maintains good detection accuracy while exhibiting strong background suppression capability.

Fig. 4 illustrates the target–background separability maps of different methods across the three datasets. For the proposed DFTD method, on the Houston2013 dataset, the green

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TABLE II
TRAINING AND TESTING TIME (IN SECONDS) OF DL-BASED METHODS FOR THREE DATASETS

	Dataset	LCNN-CD	SSDCTD	TSTTD	OS-VAE	GSSTD	CS-TTD	DFTD
Training /Fine-tuning	Houston2013	3.8913	61.4083	5330.3573	4540.8887	**0.5086**	7720.4429	66.3623
	Muufl	6.0561	6.0555	246.8655	445.7769	**0.5333**	1225.5488	61.6723
	SZUTreeData-R2	3.4055	61.2378	4905.0141	6579.7702	**0.2429**	17727.5901	64.2395
Testing	Houston2013	**0.0025**	0.1263	51.6160	2.9593	3.6945	4.5946	9.4267
	Muufl	**0.0018**	0.0033	3.5221	0.3736	0.5500	2.0632	1.0813
	SZUTreeData-R2	**0.0023**	0.1430	48.6421	2.4589	5.2557	4.5461	13.3157

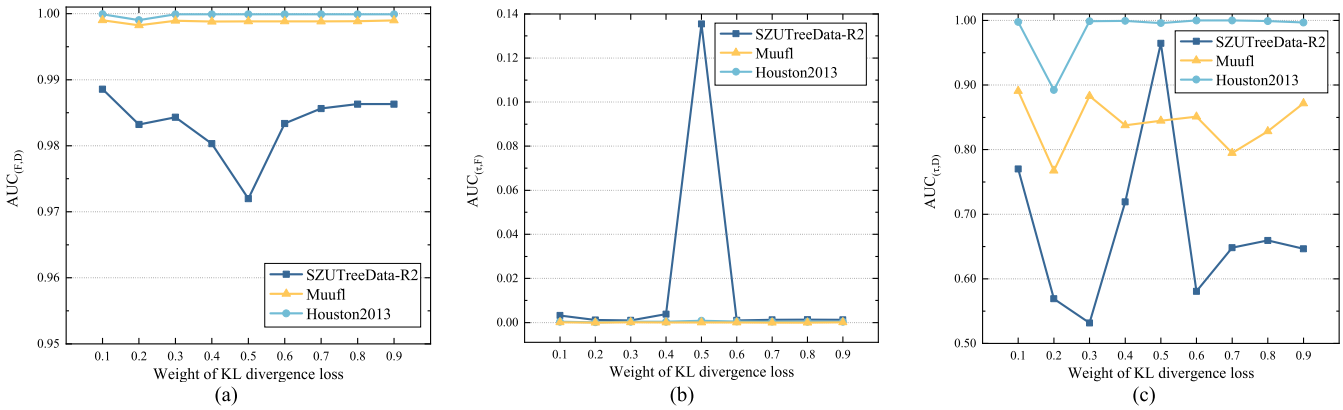


Fig. 5. Parameter analysis of the KL divergence loss weight on three datasets. (a) $\mathrm{AUC}_{(F,D)}$, (b) $\mathrm{AUC}_{(\tau,F)}$, and (c) $\mathrm{AUC}_{(\tau,D)}$.

box (representing the targets) concentrates near 1.0, while the orange box (representing the backgrounds) is completely suppressed to near zero.

This results in outstanding separability with no overlap between the two boxes. On the Muufl dataset, the green box spans a broad range from approximately 0.3 to 1.0 with the median approaching 1.0. The orange box is consistently suppressed to very low levels, creating a clear separation gap. On the SZUTreeData-R2 dataset, even though the green box shows a wide span, it still exhibits a median approaching 1.0. The orange box is effectively suppressed to near-zero levels, which allows the method to achieve reliable target discrimination. For the first two datasets, the gap between the green box and the orange box is consistently larger for the proposed method compared to other methods, indicating superior separability. Furthermore, the central line of the green box of our method is positioned higher compared to other methods, implying that DFTD better highlights the targets. The orange box is generally in a lower position, indicating that the proposed method can effectively suppress the background.

Table II presents the runtime comparison of different methods across three datasets. The lightweight approach LCNN-CD and the pretrained approach GSSTD show high training efficiency. Other comparison algorithms such as TSTTD and CS-TTD require far more time, with CS-TTD reaching up to 17 728 s on the SZUTreeData-R2 dataset. DFTD is not explicitly lightweight, yet its training time remains moderate: 66.36, 61.67, and 64.24 s on the three datasets. This is a large reduction relative to CS-TTD, and as shown in Table I, DFTD also delivers higher detection accuracy. Therefore, DFTD achieves a favorable tradeoff between accuracy and efficiency.

C. Parameter Analysis and Ablation Study

1) Parameter Analysis: This section focuses on analyzing three key parameters of the DFTD method: the weight of KL divergence loss λ_2, the number of background samples, and the weight of confidence reward λ_1. To mitigate the effects of random factors, five independent experimental runs were conducted, and the average $\mathrm{AUC}_{(F,D)}$, $\mathrm{AUC}_{(\tau,F)}$, and $\mathrm{AUC}_{(\tau,D)}$ values for three datasets are chosen.

First, the weight of KL divergence loss was set to [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]. As shown in Fig. 5, AUC values remain relatively stable on Houston2013 and Muufl datasets. AUC curves on the SZUTreeData-R2 dataset show more pronounced variations. This difference in parameter sensitivity mainly stems from the characteristics of the datasets themselves. The SZUTreeData-R2 dataset has a more complex scene structure, making the consistency constraint of cross-modal attention distributions play a more important role in this dataset. Considering overall performance across three datasets, 0.1 is selected as the optimal weight of the KL divergence loss.

Next, the number of background samples varied across [5, 10, 20, 40, 60, 80, 100, 150, 200]. As shown in Fig. 6, $\mathrm{AUC}_{(F,D)}$ values on the SZUTreeData-R2 dataset rise and then stabilize with increasing samples. Houston2013 and Muufl datasets maintain relatively stable performance. For $\mathrm{AUC}_{(\tau,F)}$, all datasets show a sharp decrease when the sample

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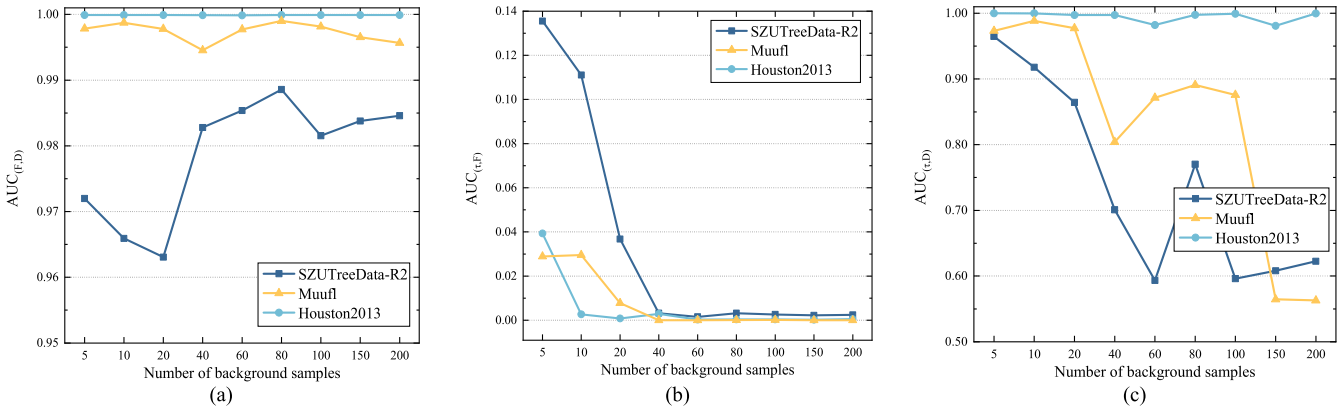


Fig. 6. Parameter analysis of background samples on three datasets. (a)AUC$_{(F,D)}$, (b) AUC$_{(\tau,F)}$, and (c) AUC$_{(\tau,D)}$.

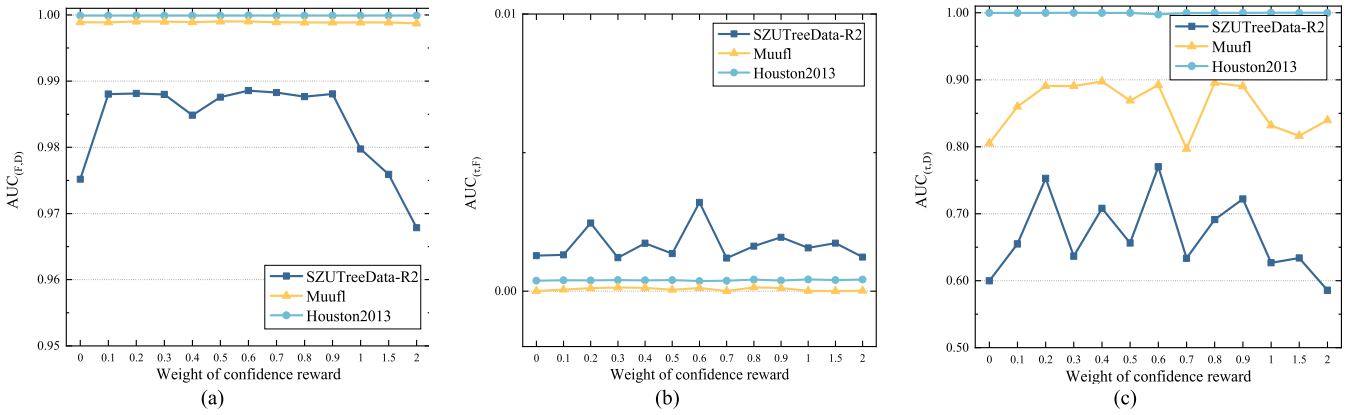


Fig. 7. Parameter analysis of the confidence reward weight on three datasets. (a) AUC$_{(F,D)}$, (b) AUC$_{(\tau,F)}$, and (c) AUC$_{(\tau,D)}$.

TABLE III
AUC VALUES OF ABLATION STUDIES FOR THREE DATASETS

Dataset		①	②	③	④
Houton2013		0.9835	0.9912	0.9998	**0.9999**
Muufl	$AUC_{(F,D)}$	0.9342	0.9944	0.9984	**0.9990**
SZUTreeData-R2		0.9169	0.9201	0.9840	**0.9886**
Houton2013		0.0054	0.0043	0.0010	**0.0004**
Muufl	$AUC_{(\tau,F)}$	0.0154	0.0126	0.0008	**0.0001**
SZUTreeData-R2		0.0568	0.0390	0.0076	**0.0032**
Houton2013		0.0398	0.8060	0.9966	**0.9976**
Muufl	$AUC_{(\tau,D)}$	0.1216	0.8223	0.8528	**0.8908**
SZUTreeData-R2		0.4355	0.3972	0.7381	**0.7702**

count reaches 40, indicating enhanced background suppression. AUC$_{(\tau,D)}$ degrades with increasing background samples. Considering all factors, 80 background samples are selected as the optimal setting.

Finally, the weight of confidence reward was set to [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2]. The confidence reward weight λ_1 controls the tradeoff between target–background separation and detection confidence in the reward function. Smaller λ_1 values guide the mask generation strategy to prioritize features with high discriminability in feature space, while larger λ_1 values shift the focus toward features that are more likely to be the target. As shown in Fig. 7, when $\lambda_1 = 0$, the model uses only a separation reward and exhibits lower performance. As λ_1 increases from 0 to 0.6, the performance of all datasets generally improves to the optimal. When λ_1 exceeds 1, the performance begins to decline, indicating that overemphasizing confidence weakens feature discriminability optimization. Based on a comprehensive analysis across three datasets, 0.6 is selected as the optimal weight of confidence reward.

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2) Ablation Study: This section evaluates the effectiveness of incorporating LiDAR data, the CFIE module, and the MaskDDPG mechanism through ablation studies. Four cases are compared:

1) using a single-branch spatial sequence Mamba encoder and a standard Transformer for HSI;

2) using a DSSM encoder and a dual-branch Transformer for HSI and LiDAR, and using a simple concatenation for feature fusion;

3) using the CFIE module and simple concatenation;

4) using the CFIE module and the CDF module.

As shown in Table III, compared to using HSI, using both HSI and LiDAR with simple concatenation can generally improve the detection performance on three datasets. For example, the $AUC_{(F,D)}$ on Houston2013, Muufl, and SZUTreeData-R2 datasets is increased by 0.0077, 0.0602, and 0.0032, respectively. Using the CFIE module can help explore the complementary information to further improve the detection performance. For example, the $AUC_{(\tau,D)}$ on Houston2013, Muufl, and SZUTreeData-R2 datasets is increased by 0.1906, 0.0305, and 0.3409, respectively. Compared to using simple concatenation, using the CDF module can generate modality-specific masks across different spatial locations to effectively fuse cross-modal encoded features and improve the detection performance.

V. Conclusion

This article proposes a DFTD method for HSI-LiDAR target detection. A cross-modal sample selection module is designed via the SID and k-means algorithms; a CFIE module is developed to extract modality-specific features via a DSSM encoder and capture cross-modal complementary information through interactive attention; and a CDF module based on MaskDDPG is designed to achieve dynamic fusion by adaptively generating modality-specific masks based on varying discriminative capacities across different spatial locations of HSI and LiDAR features under an actor–critic framework with a detection-guided reward function. Extensive experiments on three public HSI and LiDAR datasets demonstrate that the proposed DFTD method outperforms state-of-the-art approaches across multiple evaluation metrics. The method effectively suppresses background interference. However, the proposed method still has deficiencies in enhancing target discrimination. We will study how to better learn the target features with limited prior spectra.

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